1 Label-free node representation learning

Input graph
$\mathcal{V}, \mathcal{E}, \mathbf{X}$

Node features $\mathbf{X}$

Adjacency $\mathbf{A}$

	v_1	v_2	$\cdots$	v_N
v_1	0	1	$\cdots$	0
v_2	1	0	$\cdots$	1
$\cdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
v_N	0	1	$\cdots$	0

Two augmented views

View 1: edge dropping + feature masking

View 2: independently sampled

Shared graph encoder (GRACE)

Projection head contrastive training only

Frozen encoder on the original graph

View 1 projection

View 2 projection

Frozen node representation $\mathbf{G}$

2 Label-agnostic granular ball partition

Geometry-only partition in frozen representation space $\mathbf{G}$

Initialization $\mathcal{B} = \{\mathcal{V}\}$

Recursive 2-means split

Further splitting

Final granular ball partition in $\mathbf{G}$

selected ball B

Partition all nodes without using labels.

3 Leave-one-out local-consensus targets

(a) Selected granular ball B

v

v's own vote removed

(b) Leave-one-out consensus

$\mathbf{r}$ (excluding v)

$$\mathbf{q} = (1-\beta)\mathbf{r} + \beta\mathbf{e}_{i^\star}$$

If the highest vote count is tied, use $\mathbf{q} = \mathbf{r}$.

(c) Construct a soft target $\mathbf{t}$

Observed label $(1-\alpha)\mathbf{e}_z$ + Local consensus $\alpha\mathbf{q}$ = Soft target $\mathbf{t}$

$$\mathbf{t} = (1-\alpha)\mathbf{e}_z + \alpha\mathbf{q}$$

$\alpha = (C \max_i r_i - 1)/(C-1)$
Fewer than $s_{\min}$ remaining votes: retain the observed label.

4 MLP training and graph-free inference

precomputed soft target $\mathbf{t}$

MLP training with GB-LCSS targets

training node $v \in \mathcal{V}_L$

node feature $\mathbf{x}$

2-layer MLP f

softmax

prediction $\mathbf{p}$

Soft cross-entropy $-\sum_{i=1}^{C} t_i \log p_i$

Update MLP parameters

Graph-free inference (transductive)

test-node feature $\mathbf{x}$

Trained MLP f

softmax

class probabilities $\mathbf{p}$

Predicted class: 3
$\arg\max_i p_i = 3$

Observed noisy labels: C1 C2 C3 C4 C5

non-voting node

excluded self-vote

data flow

symmetric contrastive loss

Inference input: $\mathbf{x}$ only; no $\mathbf{A}$, $\mathbf{G}$, or neighbor access